实验室科研探究课第90单元

单原子层碳晶体
石墨烯
Single-layer crystalline carbon
Graphene

朱宏伟

纳米实验室

清华大学机械工程系



The Nobel Prize in Physics 2010

Andre Geim, Konstantin Novoselov

- **The Nobel Prize in Physics 2010**
  - Andre Geim
  - Konstantin Novoselov



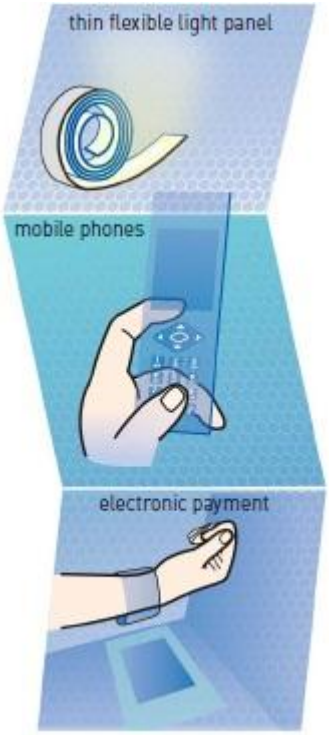
Photo: Sergeom, Wikimedia Commons

Andre Geim



Photo: University of Manchester, UK

Konstantin Novoselov





The Nobel Prize in Physics 2010 was awarded jointly to Andre Geim and Konstantin Novoselov *"for groundbreaking experiments regarding the two-dimensional material graphene"*

二维材料石墨烯的突破性实验



未来的材料

- Pencil + sticky tape = desktop supercollider + post-silicon processors

  (铅笔+胶带 = 桌面超级对撞机+后硅处理器)

- Electron superhighway: can it overtake silicon as the essential ingredient of computer chips

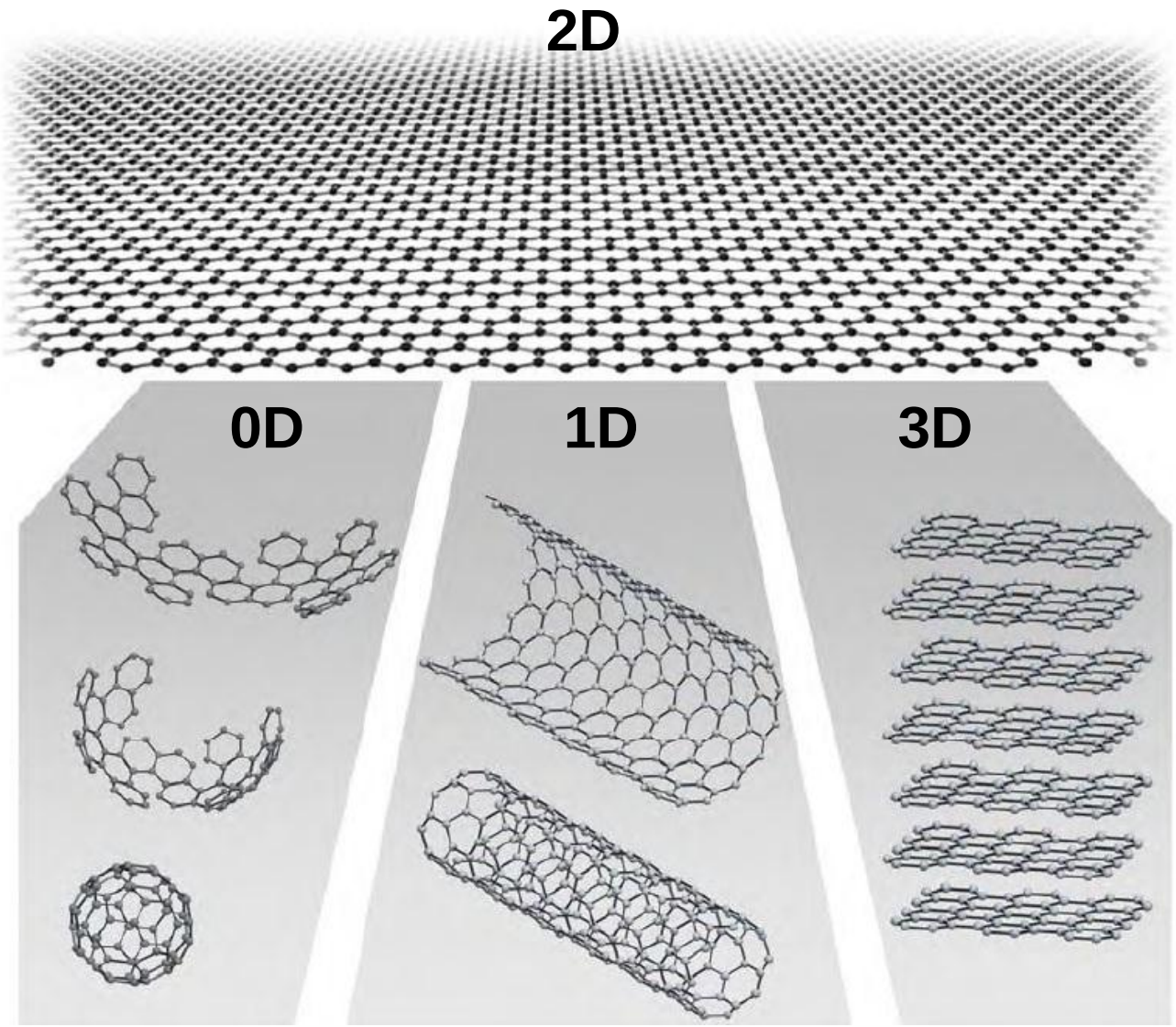
  (电子高速公路: 取代硅而成为计算机芯片的集成材料)

- Atom-thick carbon sheets set new strength record

  (单原子碳层片打破强度纪录)



石墨烯 (Graphene)



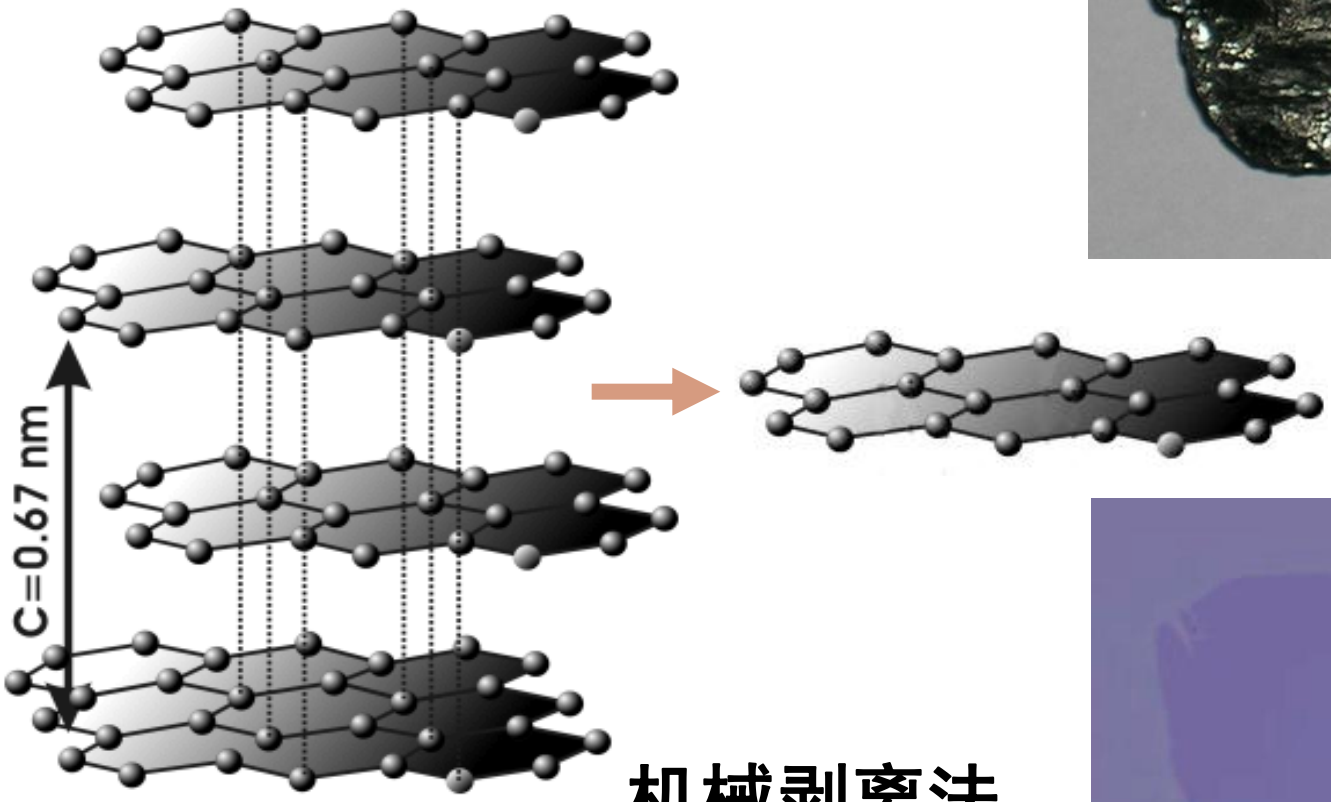
本课内容

一、石墨烯的发现、结构与特性

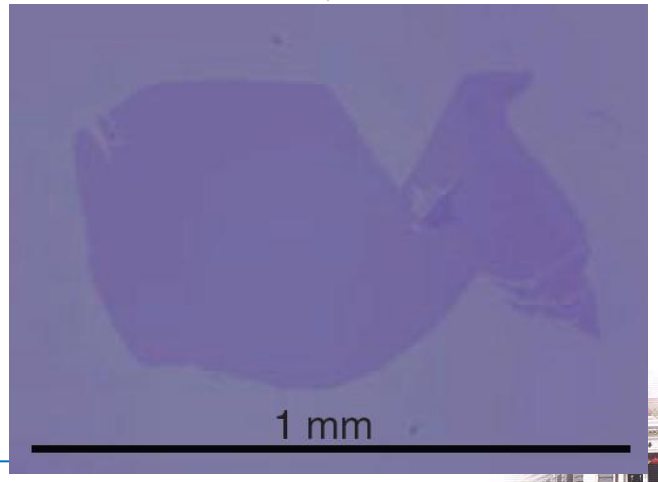
二、石墨烯的合成方法

三、实验室参观

石墨烯的发现



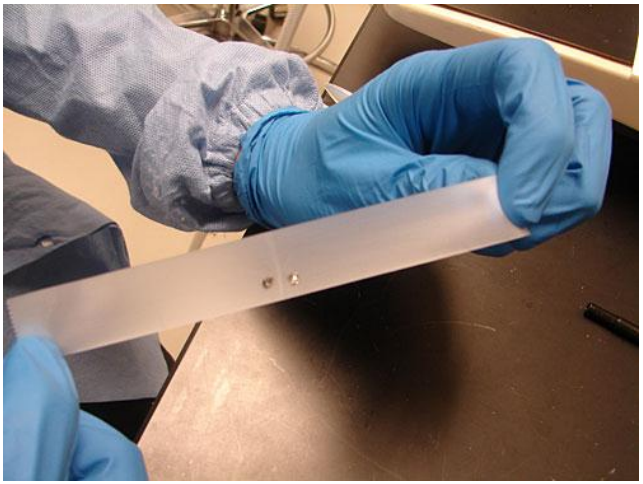


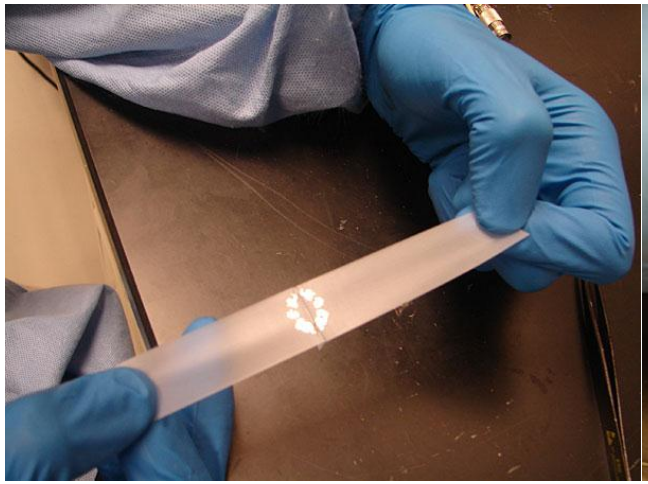


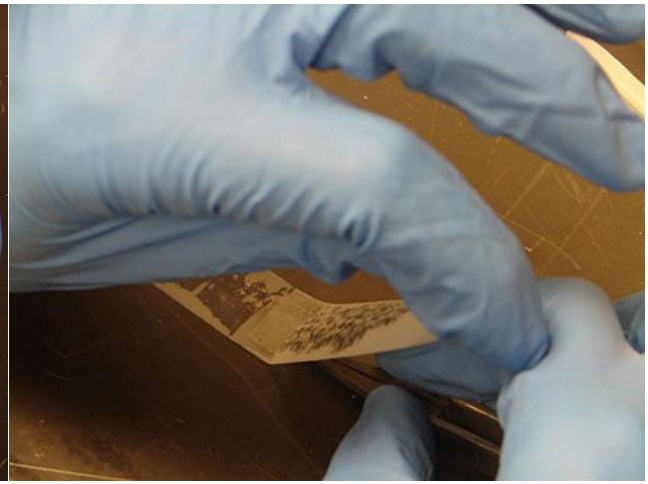
机械剥离法

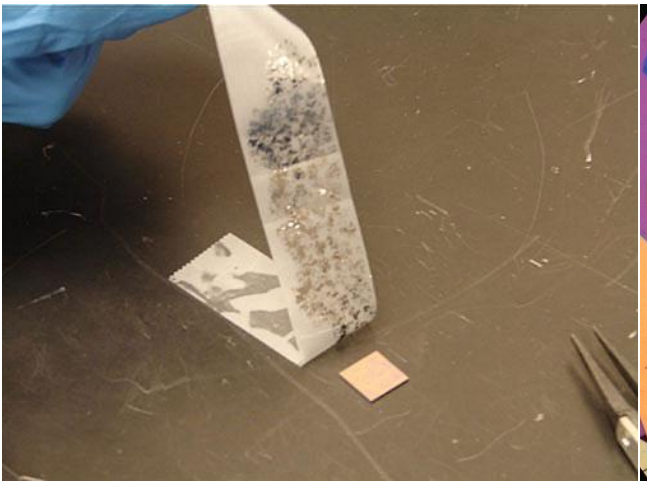
Science (2004)

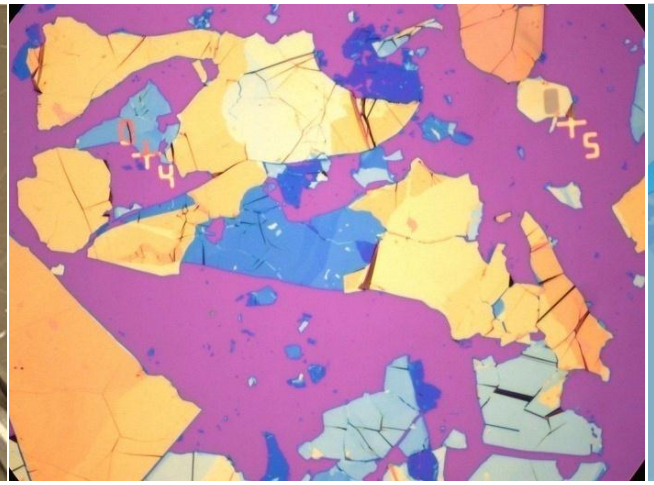
神奇的胶带

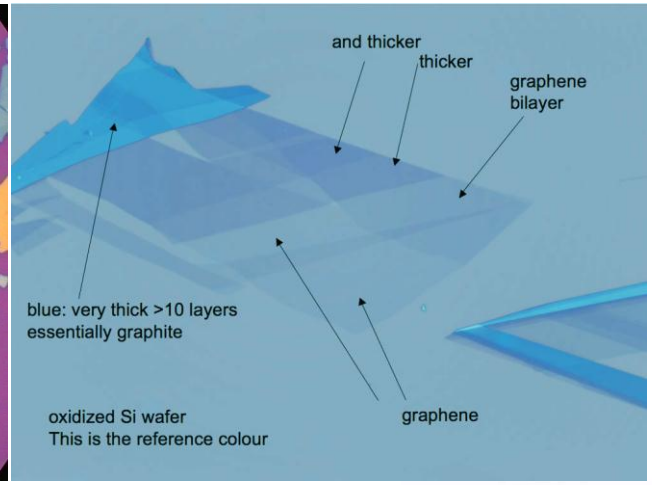








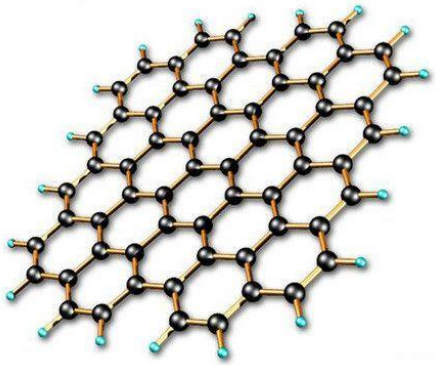



and thicker
thicker
graphene
bilayer
blue: very thick >10 layers
essentially graphite
oxidized Si wafer
This is the reference colour
graphene

Novoselov的演示

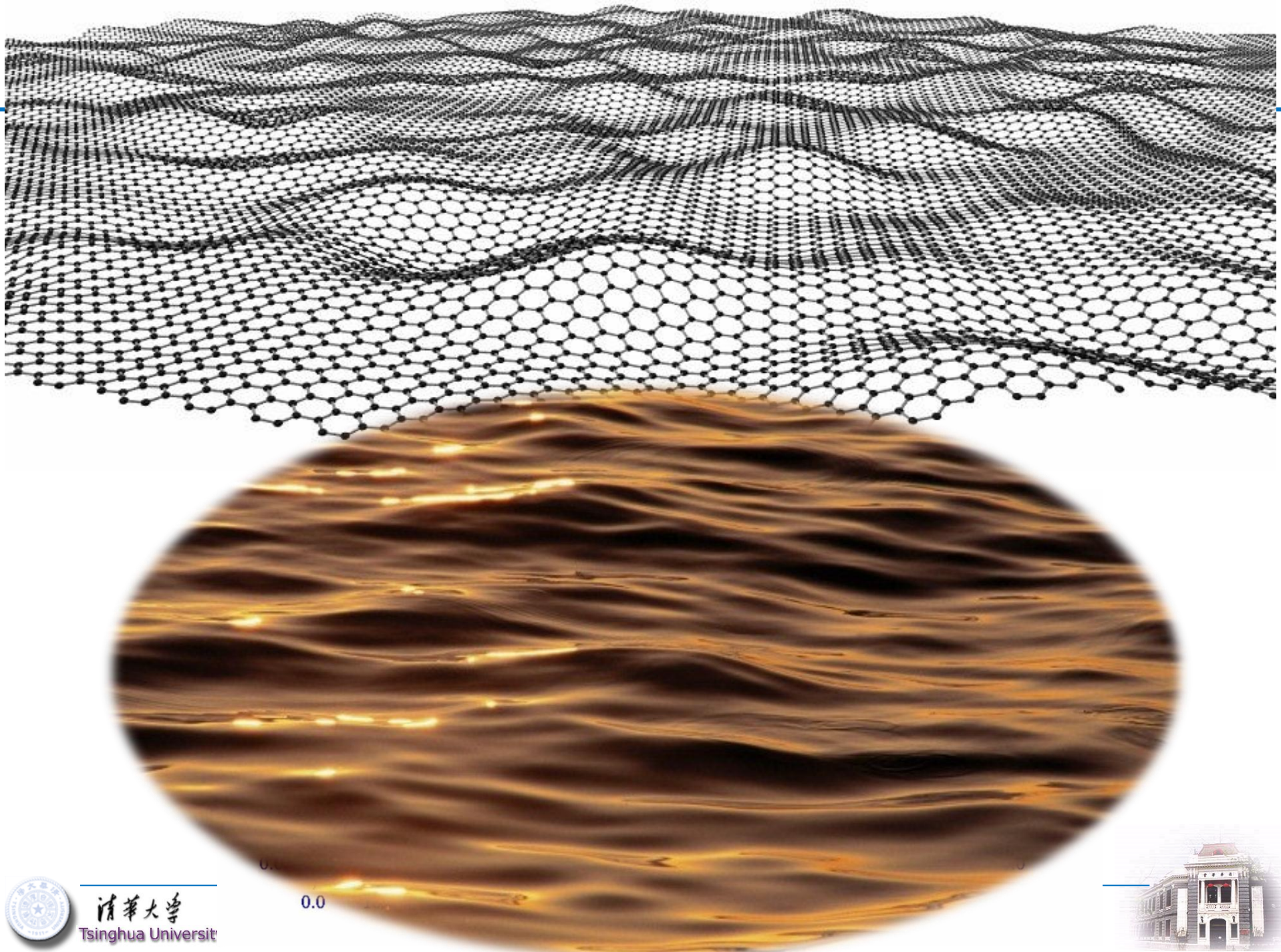


你们都错了吗？



= 颠覆传统理论

- 早在1934年，Peierls就提出准二维晶体材料由于其本身的热力学不稳定性，在常温常压下会迅速分解；
- 1966年，Mermin和Wagner提出Mermin-Wagner理论，指出表面起伏会破坏二维晶体的长程有序。

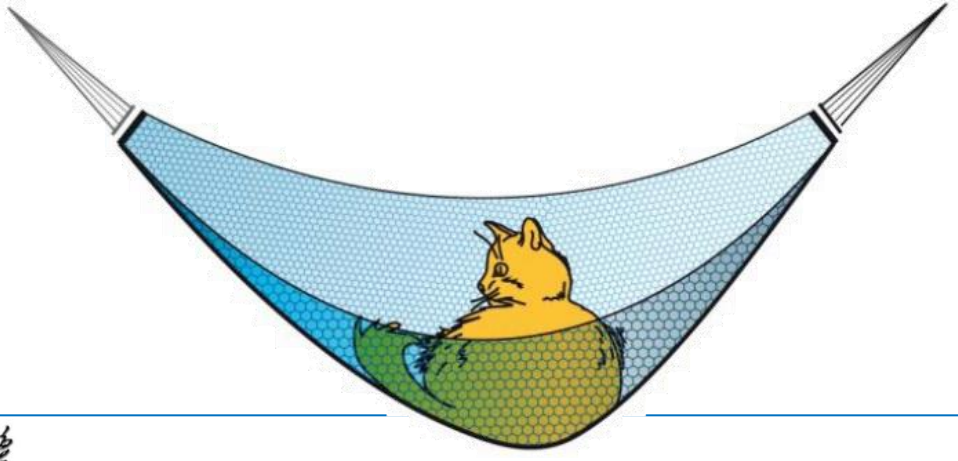

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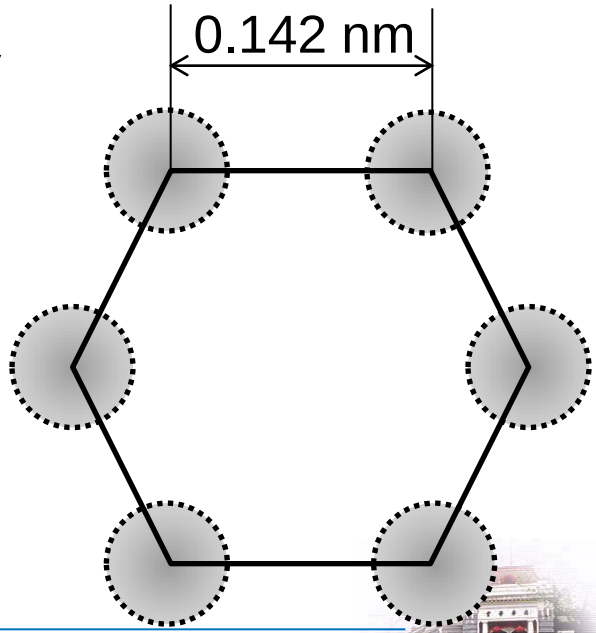
石墨烯的基本性质

- 石墨烯是已知材料中强度、导热性、导电性最高的晶体结构

石墨烯的密度：提取石墨烯中的一个正六边形碳环作为结构单元（图1），由于每个碳原子仅有1/3属于这个六边形，因此一个结构单元中的碳原子数为2。六边形的面积为0.052 nm^2。由此可计算出石墨烯的面密度为0.77 mg/m^2。假设有一张面积为1 m^2的石墨烯吊床，其质量仅为0.77 mg。

石墨烯的透光性和颜色：石墨烯的可见光透过率为97.7%，且与波长无关。因此自由悬浮的石墨烯是高度透明且无色（无味）的。

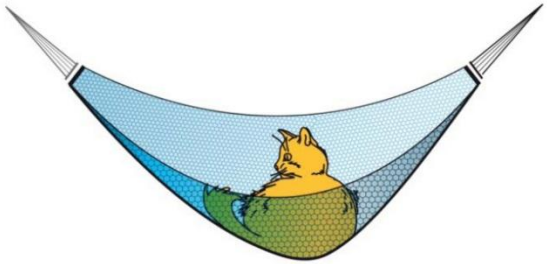




石墨烯的基本性质

石墨烯的强度：石墨烯的强度极限（抗拉强度）为42 N/m。普通钢的强度极限在250~1200 MPa范围内，即0.25~1.2×10^9 N/m^2。如果钢具有同石墨烯一样的厚度（~0.335 nm），则可推算出其二维强度极限为0.084~0.40 N/m。由此可知，石墨烯的强度约为钢的100倍。

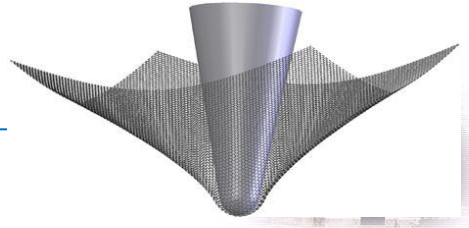
面积为1 m^2的石墨烯层片可承受4 kg的质量。也就是说，前面提到的那张吊床不仅是近乎透明的，还可以承受一只猫的重量。而这张吊床的重量仅和猫的一根胡须的重量相当。





石墨烯的电导率：石墨烯的面电阻约为31 Ω/sq，即面积为1 m^2的石墨烯吊床的电阻仅为31 Ω。

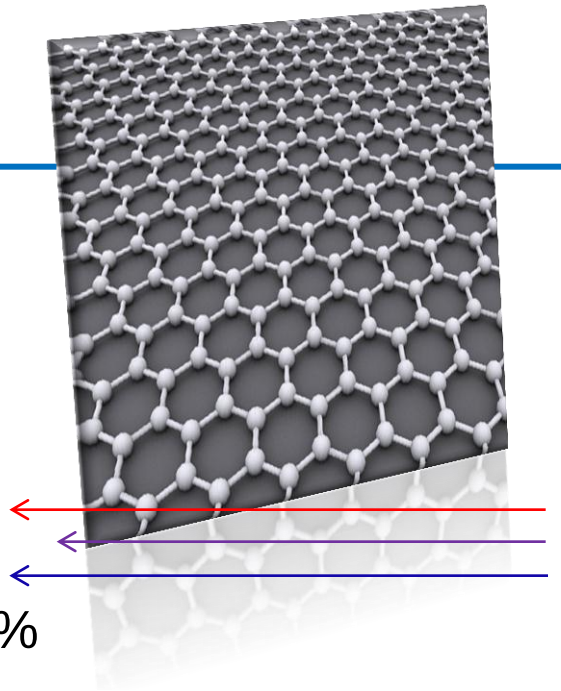
石墨烯的热导率：石墨烯的热导率实验值约为5000 $Wm^{-1}K^{-1}$，是室温下铜的热导率（401 $Wm^{-1}K^{-1}$）的10倍多。



石墨烯的哲学问题

量变引起质变 & 矛盾的统一体

- 金属vs.有效质量为零的狄拉克费米子
- 最强vs.最软
- 透明vs.不透明



Physics

Physics **2**, 30 (2009)

Pauling's dreams for graphene 鲍林

- 导体vs.绝缘体

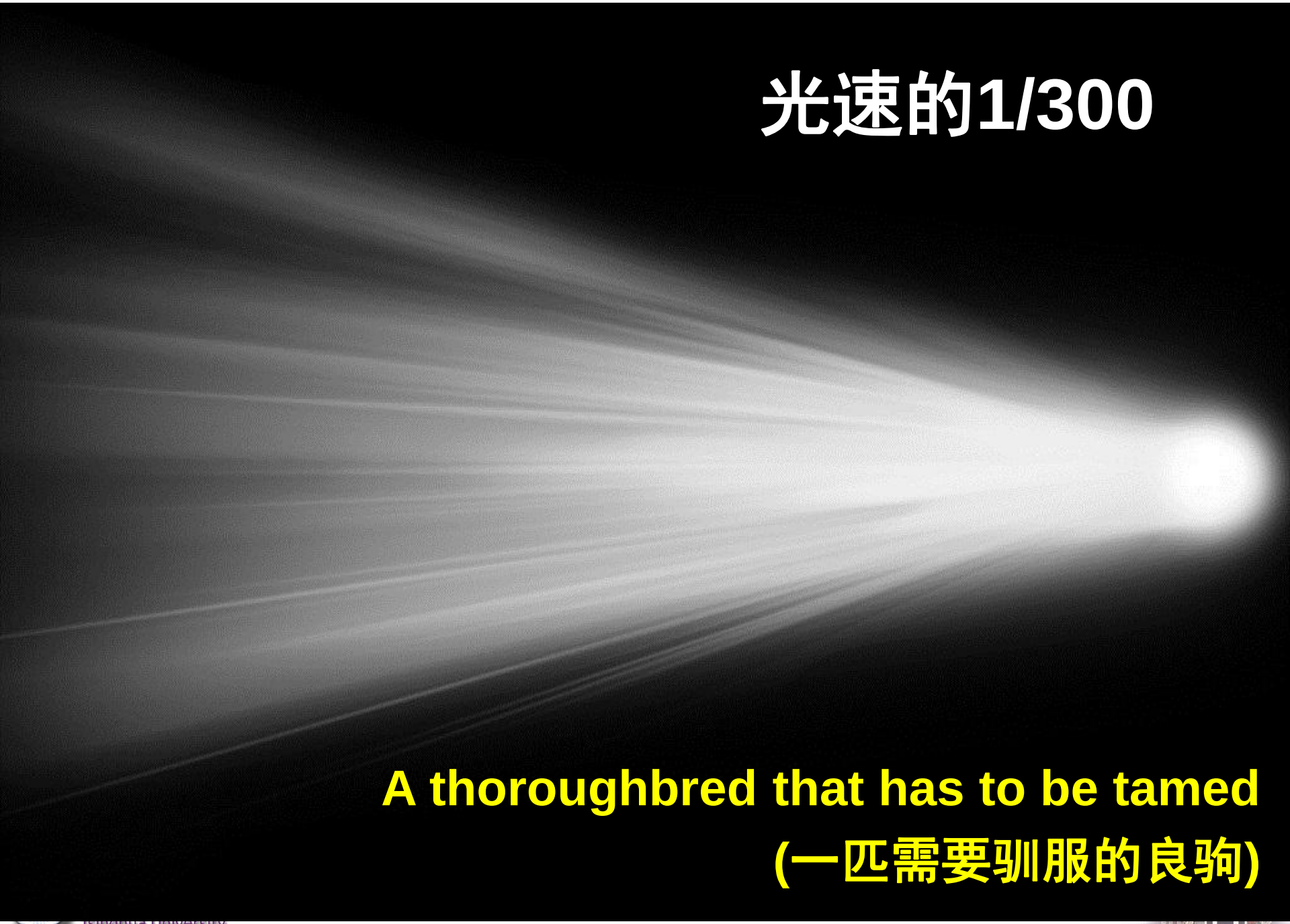
Antonio H. Castro Neto

L. Pauling proposed that graphene should be an insulator due to strong electron-electron interactions, what is today called a Mott insulator.

-The Nature of the Chemical Bond and the Structure of Molecules and Crystals: An Introduction to Modern Structural Chemistry. Cornell University Press, Ithaca, NY, 1960.

P. R. Wallace proposed, based on a theory that did not consider any electron-electron interactions, that graphene should be a semimetal.

-Phys. Rev. 1947, 71, 622.


光速的1/300
A thoroughbred that has to be tamed
(一匹需要驯服的良驹)

石墨烯物理

电子传输速度达到光速的1/300



Paul Dirac



Felix Bloch



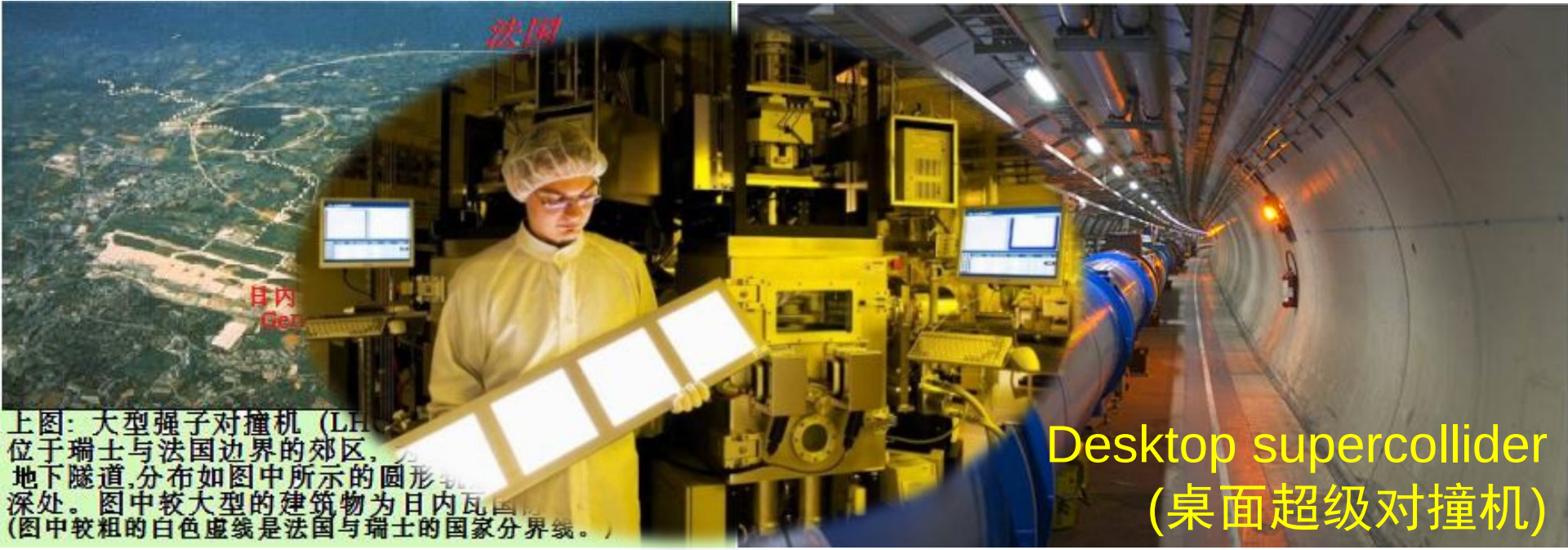
Paul Drude



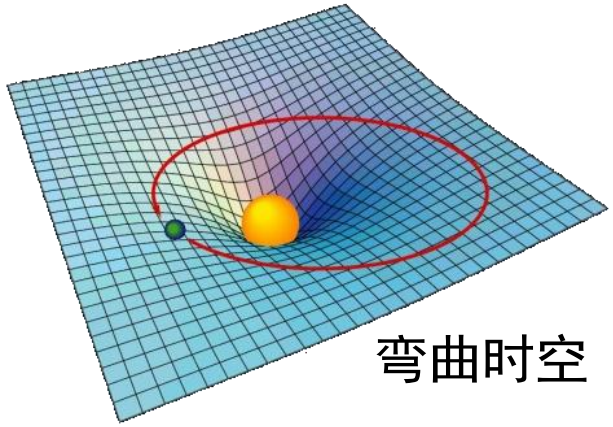
Arnold Sommerfeld



Lev Landau

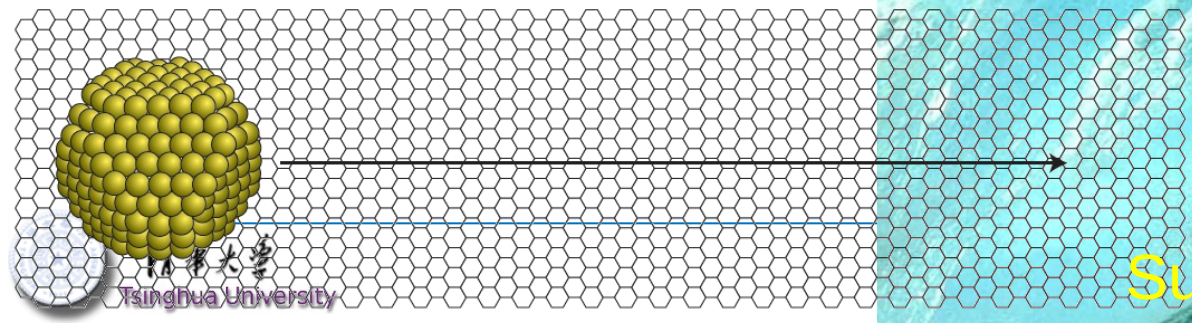


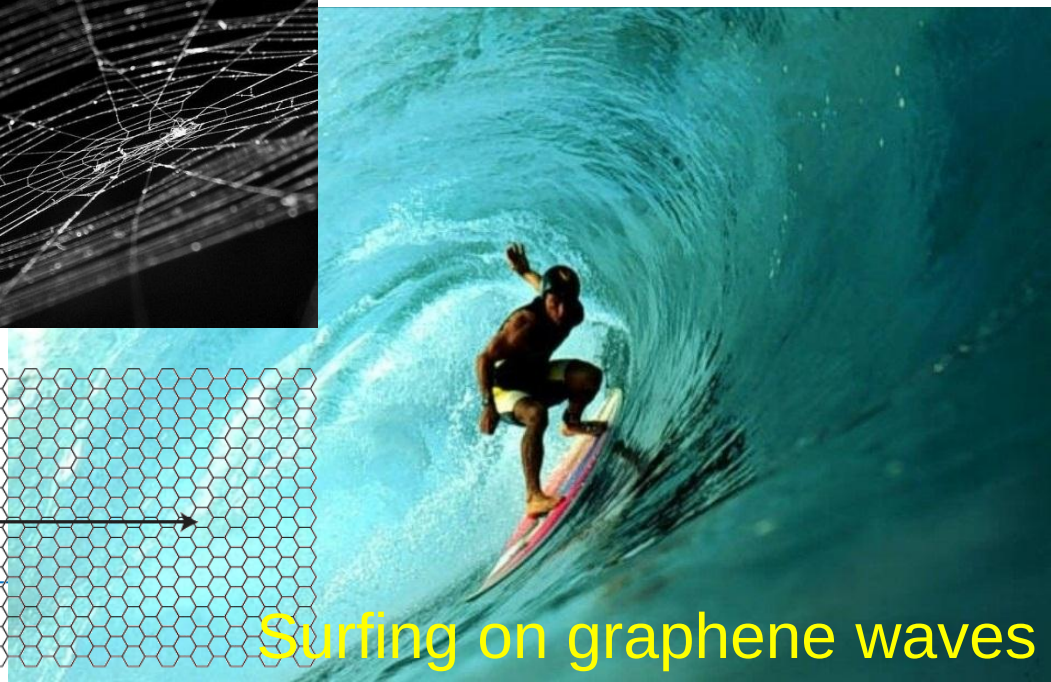
Desktop supercollider
(桌面超级对撞机)



弯曲时空







Surfing on graphene waves

石墨烯晶体管：超越硅的极限

挑战摩尔定律

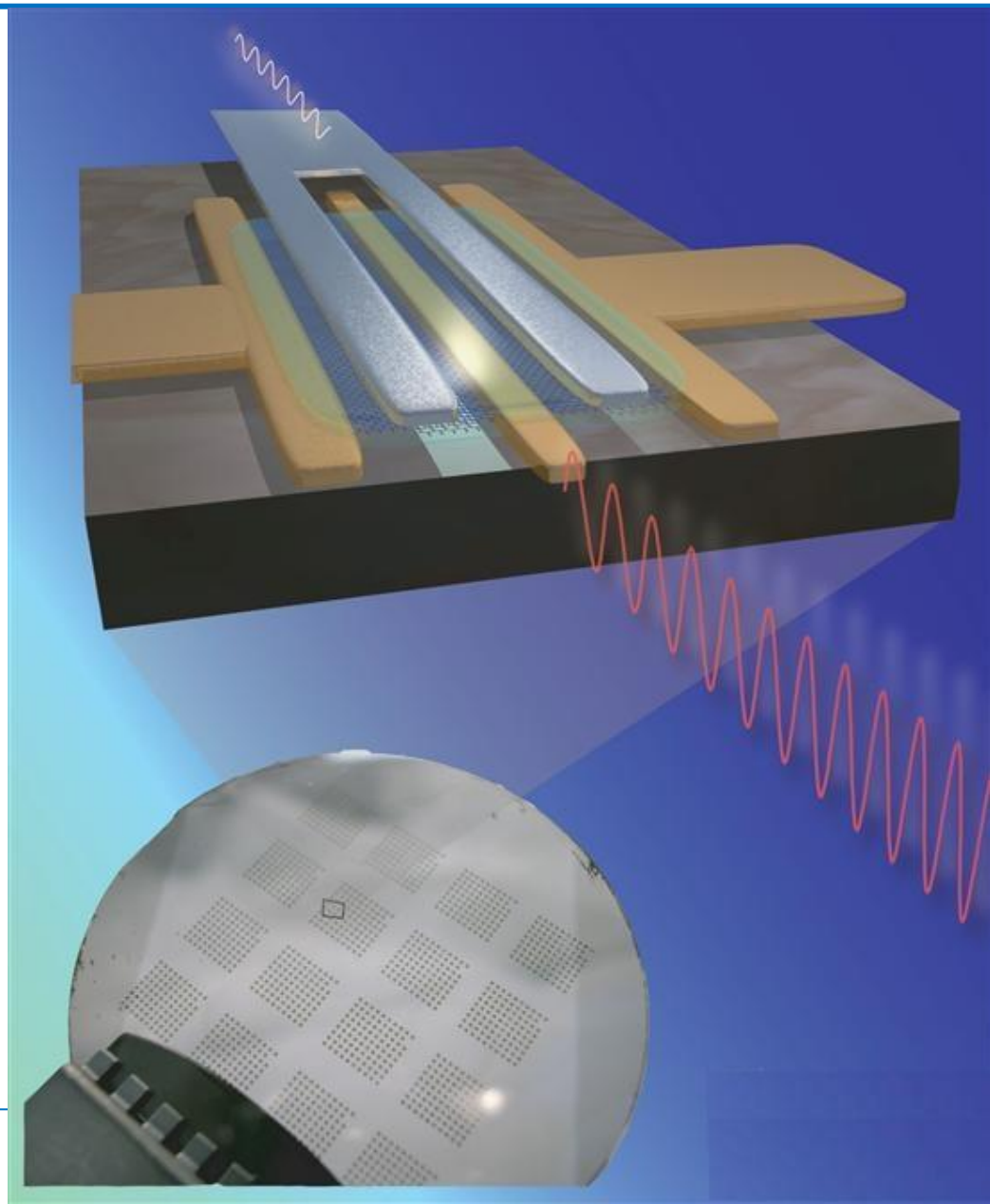
石墨烯晶体管的截止频率达到**100GHz**。

速度更快

尺寸更小

噪音更低

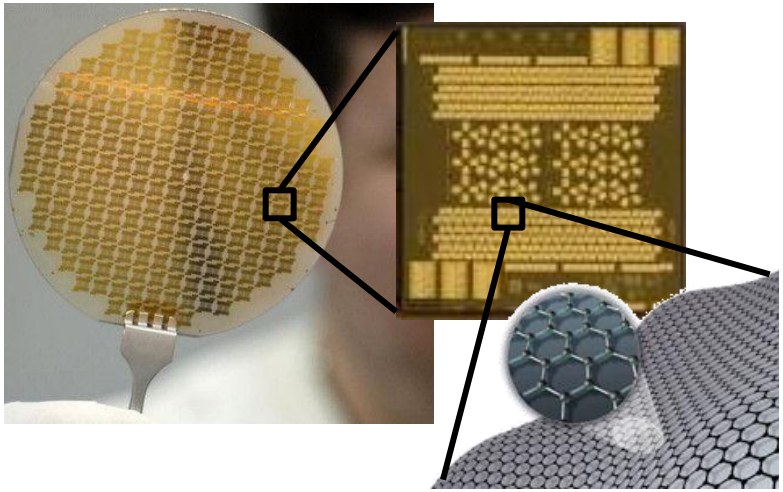
注：微处理器中的硅晶体管在极限超频状态下仅**8GHz**。最先进的硅晶体管为**40GHz**。

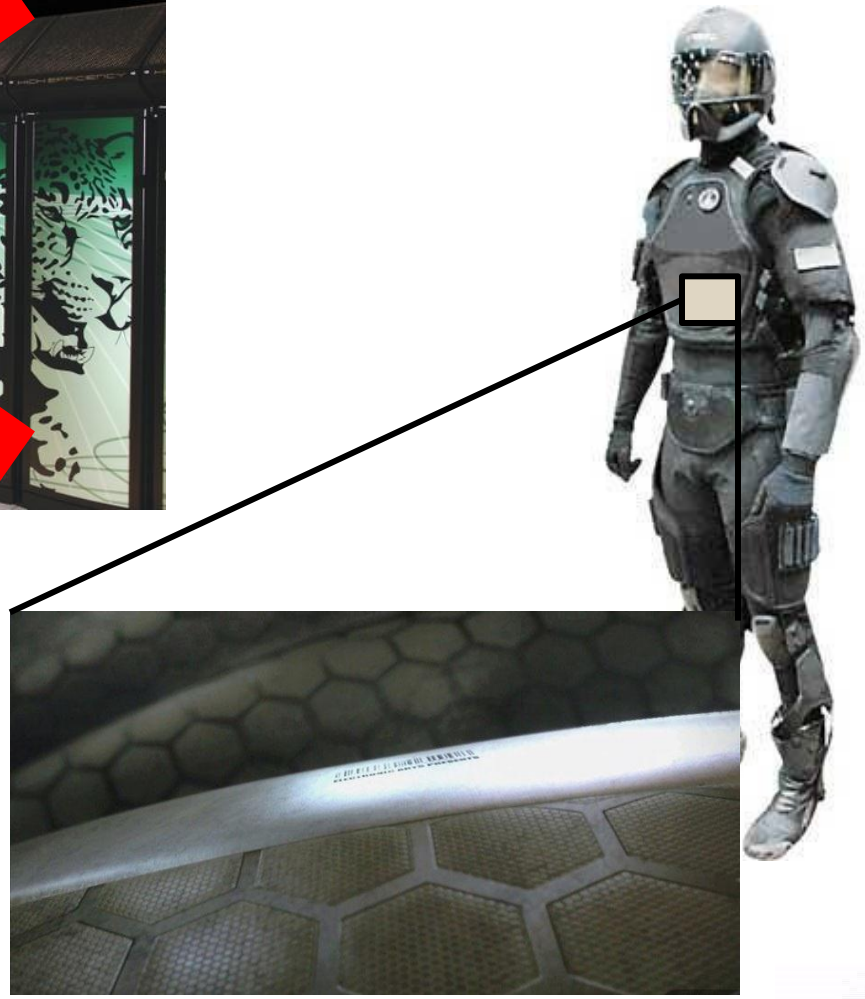


超级计算机

超强防弹衣







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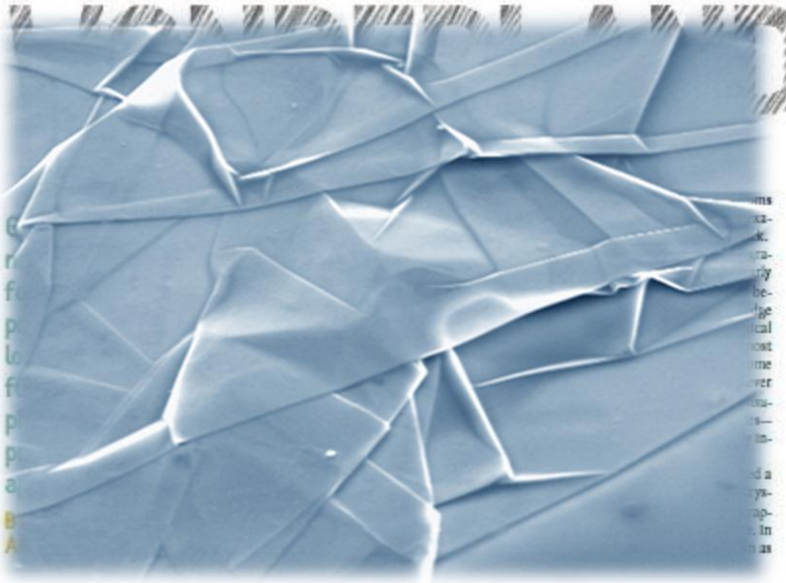
二、石墨烯的合成方法

三、实验室参观

铅笔里的石墨烯

MATERIALS

CARBON





EVERYDAY PENCIL MARKS include minute quantities of graphene, one of the hottest "new" materials in science and engineering.

micromechanical cleavage, worked surprisingly well. Investigators managed to peel off graphite films made up of fewer than 100 atomic planes. By 1990, for example, German physicists at RWTH Aachen University had isolated graphite films thin enough to be optically transparent.

A decade later one of us (Kim), working with Yuanbo Zhang, then a graduate student at Columbia University, refined the micromechanical cleavage method to create a high-tech version of the pencil—a "nanopencil," of course. "Writing" with the nanopencil yielded slices of graphite just a few tens of atomic layers thick [*see box on page 93*]. Still, the resulting material was thin graphite, not graphene. No one really expected that such a material could exist in nature.

That pessimistic assumption was put to rest in 2004. One of us (Geim), in collaboration with then postdoctoral associate Kostya S. Novoselov and his co-workers at the University of Manchester in England, was studying a variety of approaches to making even thinner samples of graphite. At that time, most laboratories began such attempts with soot, but Geim and his colleagues serendipitously started with bits of debris left over after splitting graphite by brute force. They simply stuck a flake of graphite debris onto plastic adhesive tape, folded the sticky side of the tape over the flake and then pulled the tape apart, cleaving the flake in two. As the experimenters repeated the process, the resulting fragments grew thinner [*see box on page 95*]. Once the investigators had many thin fragments, they meticulously examined the pieces—and were astonished to find that some were only one atom thick. Even more unexpectedly, the newly identified bits of graphene turned out to have high crystal quality and to be chemically stable even at room temperature.

The experimental discovery of graphene led to a deluge of international research interest. Not only is it the thinnest of all possible materials, it is also extremely strong and stiff. Moreover, in its pure form it conducts electrons faster at room temperature than any other substance. Engineers at laboratories worldwide are currently scrutinizing the stuff to determine whether it can be fabricated into products such as supertough composites, smart displays, ultrafast transistors and quantum-dot computers.

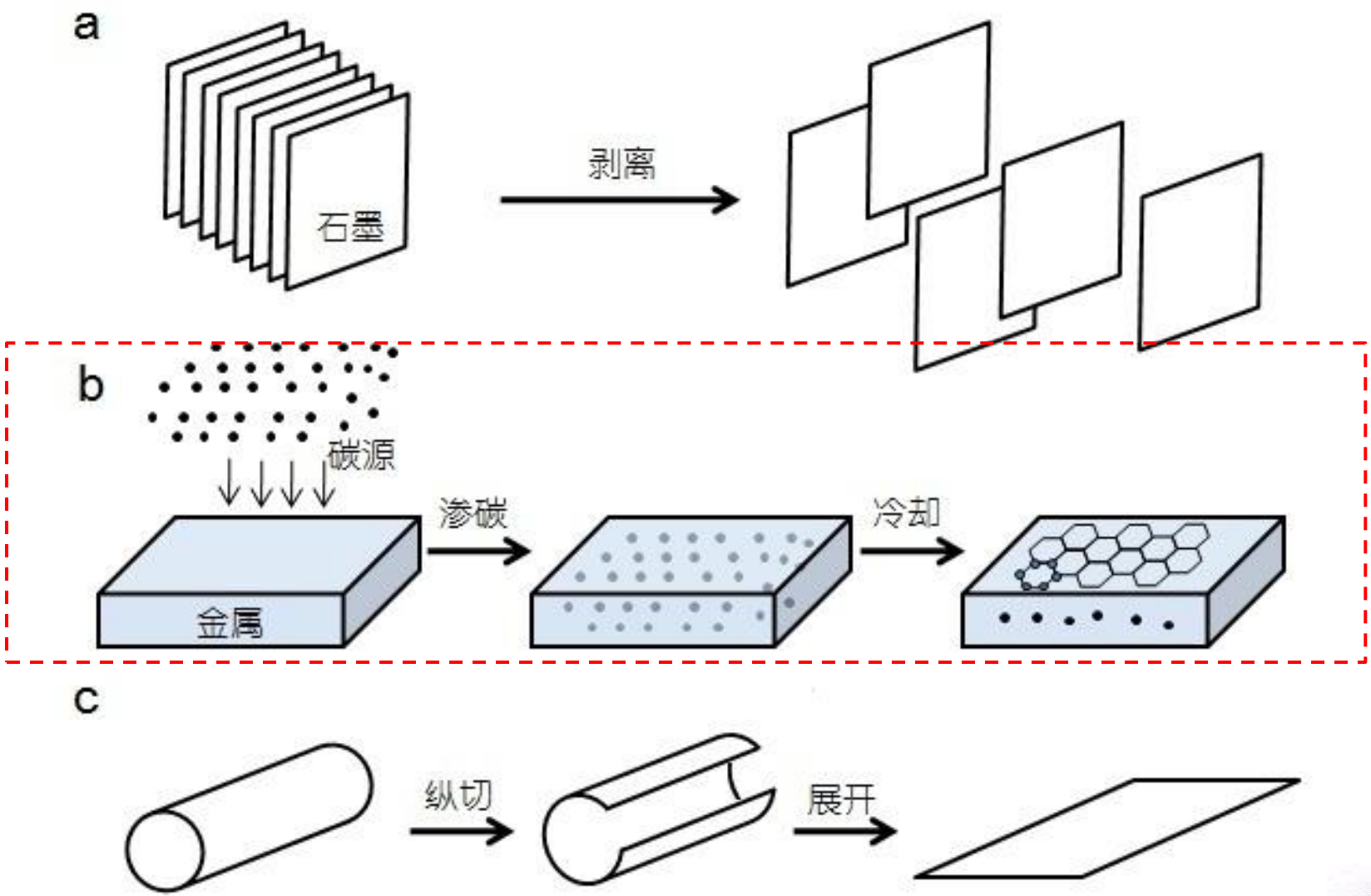
In the meantime, the peculiar nature of graphene at the atomic scale is enabling physicists to delve into phenomena that must be described by relativistic quantum physics. Investigating such

KEY CONCEPTS

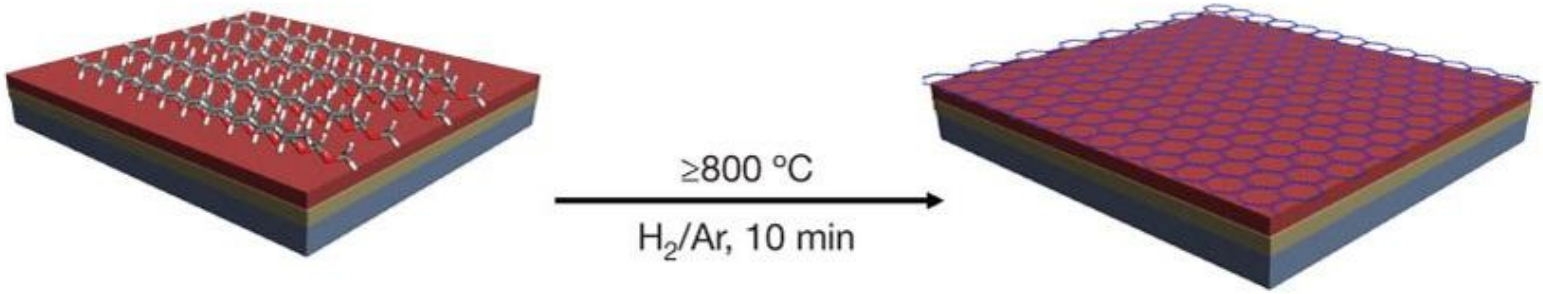
- Graphene is a one-atom-thick sheet of carbon that stacks with other such sheets to form graphite—pencil "lead." Physicists have only recently isolated the material.
- The pure, flawless crystal conducts electricity faster at room temperature than any other substance.
- Engineers envision a range of products made of graphene, such as ultrahigh-speed transistors. Physicists are finding the material enables them to test a theory of exotic phenomena previously thought to be observable only in black holes and high-energy particle accelerators.

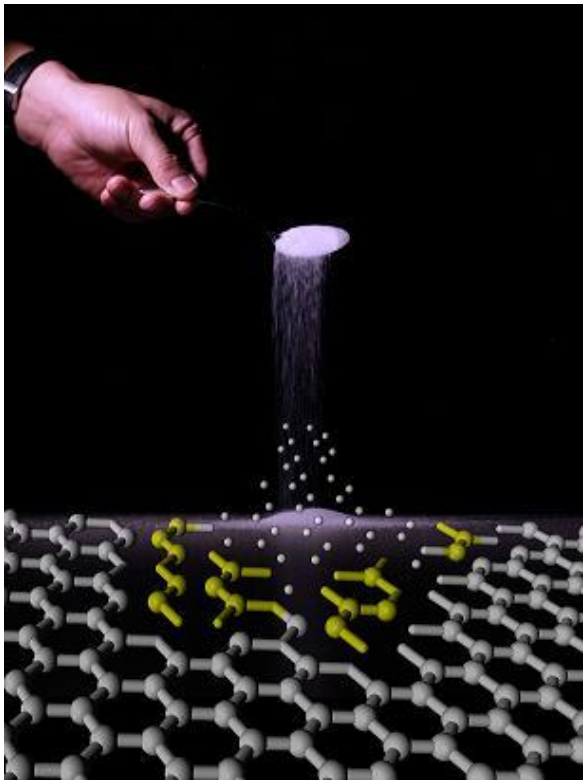
—*The Editors*

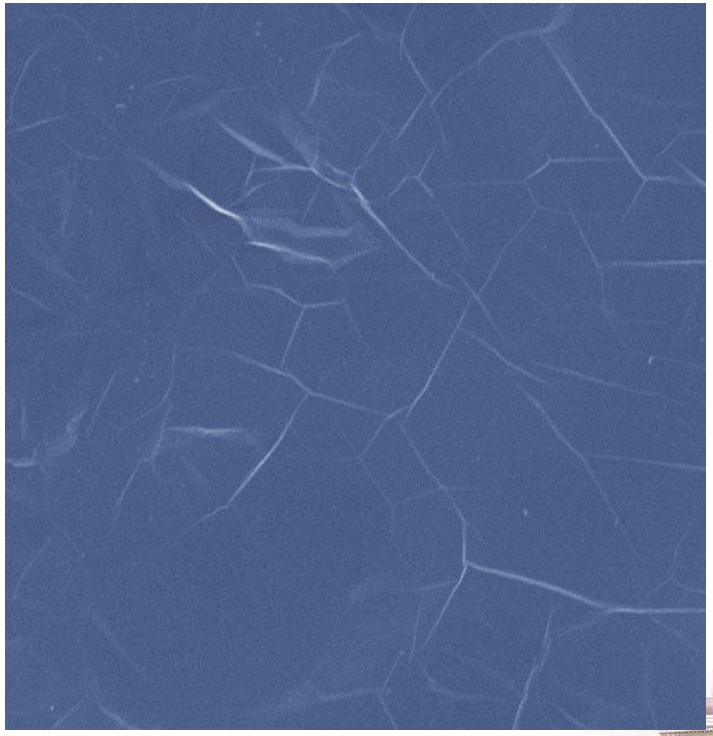
石墨烯的合成方法



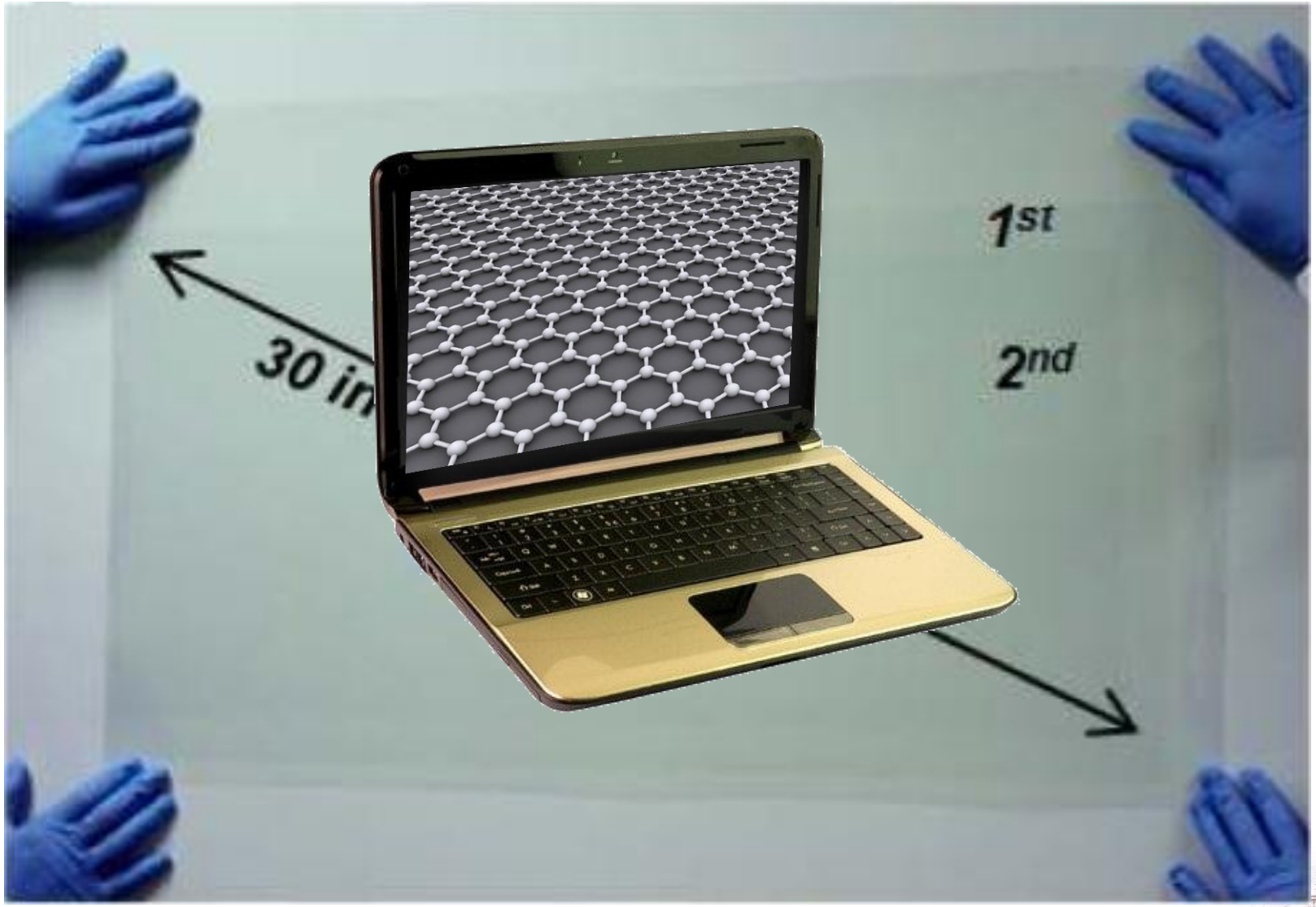
从蔗糖到石墨烯







大面积合成

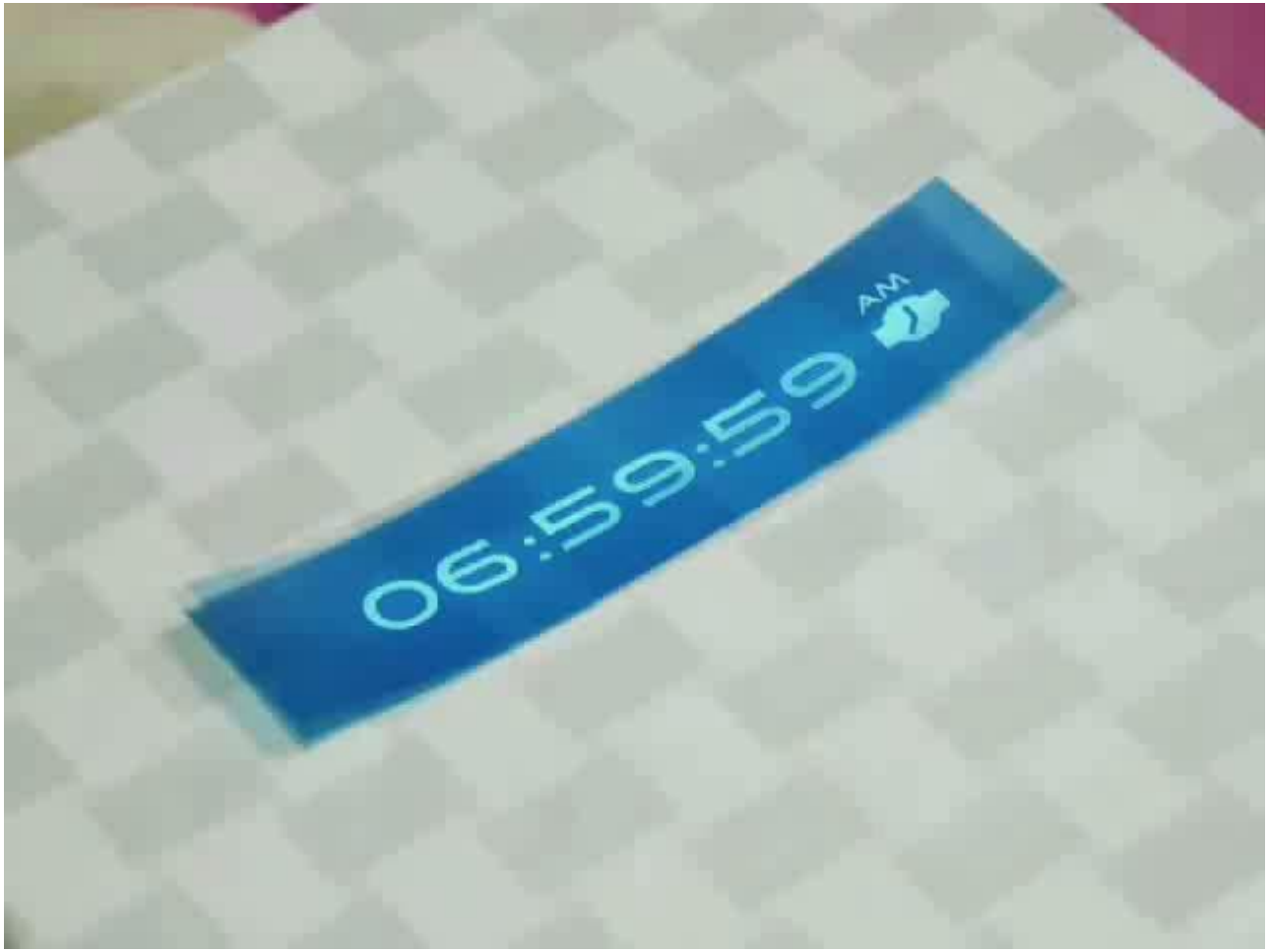


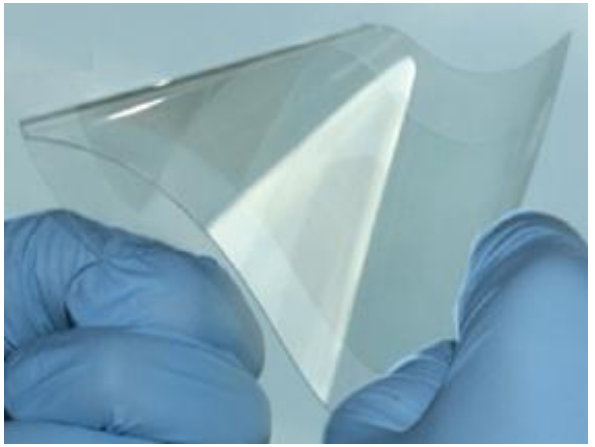
薄膜应用

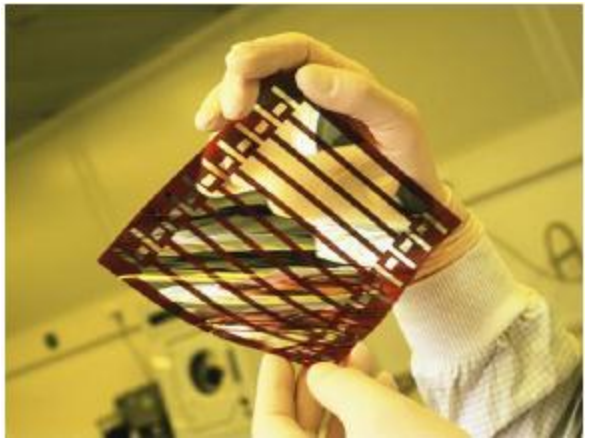

Industrial & Home Networking
Medical Electronic
Network & Server Equipments
Automotive Car Electronic
Mil & Aerospace
Next Gen. Gaming Devices / Stations
Lower Volumes
> 1M units
MOTIVATIONS:
• Smaller Size & Lighter Weight
• Innovative Function & Flexible Design
• Integrated System
• High Electrical Performance
• Low Cost & Time to Market
> 10M units
Mobile phones
LED lighting
> 1B units
> 100M units
High Volume
Set-Top Box
PMP
HDTV
DSC
Netbook & MID
3G Data Card


未来显示技术
MPlayer
Ordenador
2:55:43

柔性器件







本课内容

一、石墨烯的发现、结构与特性

二、石墨烯的合成方法

三、实验室参观